

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 11A.5

ACTIVATION ENERGY AND CATALYSIS

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 11: KINETICS 2
SUB-TOPIC:	11A.5: Activation Energy and Catalysis
TOTAL QUESTIONS:	4 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	9 Marks
EXAMINER FAQs:	3 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 11A.5

Q1. Q1. Explain how a catalyst increases the rate of a chemical reaction.

[2 Marks]

WCH14/01/Oct22/Q3(a)

Q2. Q2. Distinguish between homogeneous and heterogeneous catalysis.

[2 Marks]

WCH14/01/Oct22/Q3(b)

Q3. Q3. Describe the steps involved in heterogeneous catalysis.

[3 Marks]

WCH14/01/Jun21/Q4(a)

Q4. Q4. Using the Maxwell-Boltzmann distribution, explain why a small increase in temperature leads to a large increase in the rate of reaction.

[2 Marks]

WCH14/01/Jun21/Q4(b)

SECTION B: WORKED MARK SCHEME — 11A.5

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: Q1. Explain how a catalyst increases the rate of a chemical reaction. • Provides an alternative pathway/mechanism (1) • With a lower activation energy (1)	[2]
Q2	Q2: Q2. Distinguish between homogeneous and heterogeneous catalysis. • Homogeneous: catalyst is in the same phase/state as the reactants (1) • Heterogeneous: catalyst is in a different phase/state from the reactants (1)	[2]
Q3	Q3: Q3. Describe the steps involved in heterogeneous catalysis. • Adsorption of reactants onto the catalyst surface (1) • Bonds break and form / reaction occurs on the surface (1) • Desorption of products from the surface (1)	[3]
Q4	Q4: Q4. Using the Maxwell-Boltzmann distribution, explain why a small increase in temperature leads to a large increase in the rate of reaction. • Many more molecules have energy greater than or equal to the activation energy (1) • Higher frequency of successful collisions (1)	[2]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 11A.5**FAQ #1: Does a catalyst change the position of equilibrium?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Does a catalyst change the position of equilibrium?

MODEL ANSWER / EXAMINER EXPECTATION:

No, it increases the rate of both the forward and backward reactions equally, so the equilibrium position remains unchanged.

FAQ #2: What is catalyst poisoning?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

What is catalyst poisoning?

MODEL ANSWER / EXAMINER EXPECTATION:

When impurities bind strongly to the active sites of a heterogeneous catalyst, blocking reactants from adsorbing and reducing catalytic efficiency.

FAQ #3: Why is adsorption strength important for a heterogeneous catalyst?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Why is adsorption strength important for a heterogeneous catalyst?

MODEL ANSWER / EXAMINER EXPECTATION:

It must be strong enough to weaken bonds in the reactants, but weak enough to allow products to desorb.